EVOLUTIONARY COMPUTATION

LABORATORY 02

Greedy heuristics

for the Traveling Salesperson Problem with Node Costs

ASSIGNMENT 2 - REPORT

SOURCE REPOSITORY

https://github.com/Flat-Earther/Evolutionary-Computation/tree/main/lab-02-greedy-regret-heuristics

Authors

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OBJECTIVE

Minimize cycle length + selected node costs

Choose exactly ceil(n / 2) nodes and arrange them into a closed cycle.

The supplied instances contain 200 nodes, so every solution selects 100.

INSTANCES

TSPA and TSPB

200 nodes each - 100 selected

METHODS

8 greedy and regret methods

200 evaluated solutions per method

Poznań University of Technology - Artificial Intelligence

October 2026

1. Problem description

The input consists of n nodes, each with integer planar coordinates (x, y) and an integer node cost.

The task is to choose exactly k = ceil(n / 2) distinct nodes and specify their cyclic visiting order.

The cycle must return from its last selected node to its first selected node.

OBJECTIVE

f(S) = cycle length(S) + sum of costs of selected nodes

DISTANCE

d(i,j) = round(sqrt((xi-xj)^2 + (yi-yj)^2))

Distances are calculated once
and stored in a distance matrix.

DATA SEPARATION

Optimization methods access
the distance matrix and node
costs, not the original coordinates.

For the supplied TSPA and TSPB instances, n = 200 and k = 100.

This problem combines a subset-selection decision with a cyclic routing decision. The algorithms below construct feasible solutions directly and evaluate every candidate change using both distance and node cost.

Solution representation

A solution is an ordered list of k distinct node indices.

The closing edge from the last node to the first is included in the objective.

All reported indices are zero-based, matching the supplied solution checker.

2. Algorithms and pseudocode

2.1 2-regret - open path

Builds an open path and prioritizes nodes whose second-best position is much worse than their best position.

```
Algorithm Regret2Path(D, C, start)
1. S <- [start]; U <- V \ {start}
2. while |S| < k:
 for each v in U and each path gap p:
 delta(v,p) <- added distance + C[v]
 best(v), p(v) <- smallest delta(v,p)
 second(v) <- second-smallest delta(v,p)
 regret(v) <- second(v) - best(v)
 choose v with largest regret(v)
 insert v into gap p(v); remove v from U
3. close S with an edge from last to first
4. return S and objective F(S)
```

2.2 2-regret - closed cycle

Maintains a closed cycle and evaluates every possible insertion edge independently.

```
Algorithm Regret2Cycle(D, C, start)
1. v2 <- argmin over v != start of D[start][v] + C[v]
2. S <- [start, v2]; U <- V \ S
3. while |S| < k:
 for each v in U and each cycle edge (a,b):
 delta(v,a,b) <- D[a][v] + D[v][b]
 - D[a][b] + C[v]
 best(v), e(v) <- smallest delta(v,a,b)
 second(v) <- second-smallest delta(v,a,b)
 regret(v) <- second(v) - best(v)
 choose v with largest regret(v)
 insert v into edge e(v); remove v from U
4. return the closed cycle S and F(S)
```

2.3 Weighted regret - open path

Combines open-path regret with the immediate insertion cost; both weights are one in the experiment.

```
Algorithm WeightedRegretPath(D, C, start)
1. S <- [start]; U <- V \ {start}
2. wr <- 1; wd <- 1
3. while |S| < k:
 for each v in U and each path gap p:
 delta(v,p) <- added distance + C[v]
 best(v), p(v) <- smallest delta(v,p)
 second(v) <- second-smallest delta(v,p)
 regret(v) <- second(v) - best(v)
 score(v) <- wr * regret(v) - wd * best(v)
 choose v with largest score(v)
 insert v into gap p(v); remove v from U
4. close S and return S and F(S)
```

2.4 Weighted regret - closed cycle

Combines closed-cycle regret with the immediate insertion cost using explicit equal weights.

```
Algorithm WeightedRegretCycle(D, C, start)
1. v2 <- argmin over v != start of D[start][v] + C[v]
2. S <- [start, v2]; U <- V \ S
3. wr <- 1; wd <- 1
4. while |S| < k:
 for each v in U and each cycle edge (a,b):
 delta(v,a,b) <- D[a][v] + D[v][b]
 - D[a][b] + C[v]
 best(v), e(v) <- smallest delta(v,a,b)
 second(v) <- second-smallest delta(v,a,b)
 regret(v) <- second(v) - best(v)
 score(v) <- wr * regret(v) - wd * best(v)
 choose v with largest score(v)
 insert v into edge e(v); remove v from U
5. return the closed cycle S and F(S)
```

3. Computational experiment - TSPA

Each row aggregates 200 solutions generated from all possible starting nodes.

Weighted methods use equal weights: regret_weight = 1 and delta_weight = 1.

For each method, minimum, maximum, average, and standard deviation are calculated.

Experimental results - TSPA

Algorithm	Minimum	Maximum	Average	Std. dev.
Random solution	237026	285754	261754.81	9444.81
Nearest neighbor - end	84362	90075	87476.63	1328.22
Nearest neighbor - any position	72328	77022	74347.62	981.22
Greedy cycle	72590	75281	73793.41	527.45
2-regret - path	112968	131943	121241.39	3928.02
2-regret - cycle	111748	129972	120502.06	3493.64
Weighted regret - path	72307	78176	75189.33	1612.85
Weighted regret - cycle	**72011**	76027	**73335.27**	869.43

lowest minimum lowest average

Result summary

The best individual solution for TSPA was found by Weighted regret - cycle, with objective 72011.

The best average result for this instance was obtained by Weighted regret - cycle, with average objective 73335.27.

All generated routes passed the internal structural validation.

Runtime reference

Method	Runtime
Random solution	2 ms
Nearest neighbor - end	88 ms
Nearest neighbor - any position	21143 ms
Greedy cycle	11978 ms
2-regret - path	21526 ms
2-regret - cycle	12376 ms
Weighted regret - path	21631 ms
Weighted regret - cycle	12443 ms

3. Computational experiment - TSPB

Each row aggregates 200 solutions generated from all possible starting nodes.

Weighted methods use equal weights: regret_weight = 1 and delta_weight = 1.

For each method, minimum, maximum, average, and standard deviation are calculated.

Experimental results - TSPB

Algorithm	Minimum	Maximum	Average	Std. dev.
Random solution	181550	228271	206936.33	8541.75
Nearest neighbor - end	56007	63166	59397.34	2093.12
Nearest neighbor - any position	52915	66164	59465.47	2193.52
Greedy cycle	55655	63734	59649.91	2804.92
2-regret - path	67937	79102	73139.61	2178.64
2-regret - cycle	66167	77274	72327.22	1916.47
Weighted regret - path	**51301**	65594	**57804.59**	2791.11
Weighted regret - cycle	52860	63406	57903.04	3551.12

lowest minimum lowest average

Result summary

The best individual solution for TSPB was found by Weighted regret - path, with objective 51301.

The best average result for this instance was obtained by Weighted regret - path, with average objective 57804.59.

All generated routes passed the internal structural validation.

Runtime reference

Method	Runtime
Random solution	3 ms
Nearest neighbor - end	90 ms
Nearest neighbor - any position	21330 ms
Greedy cycle	12329 ms
2-regret - path	21873 ms
2-regret - cycle	12814 ms
Weighted regret - path	21845 ms
Weighted regret - cycle	12763 ms

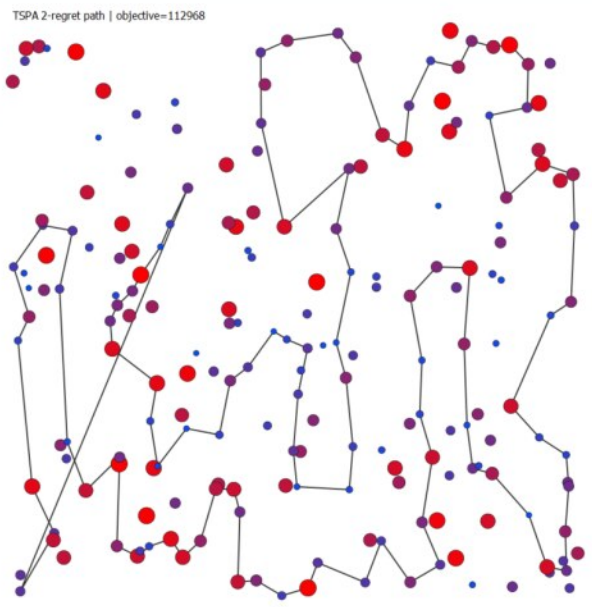
4. Best solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

2-regret - path

min=112968 max=131943 avg=121241.

best = 112968



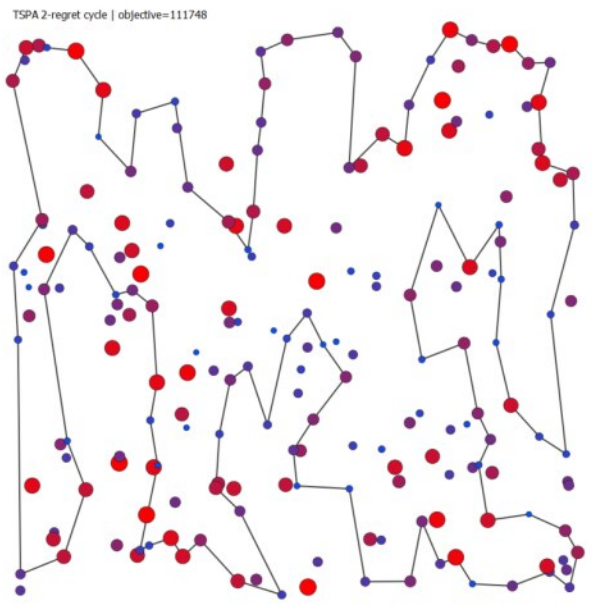
Best solution nodes:

```
4 76 132 68 90 83 160 12 23 30 187 141 87
102 185 67 144 179 140 73 135 58 84 99 36
33 143 62 190 112 115 175 39 154 34 109
186 136 24 163 19 29 17 101 178 59 155 21
147 37 47 117 130 159 170 75 193 122 107
103 42 106 81 128 142 194 174 176 0 138
182 111 158 192 181 118 114 25 22 98 70 96
10 183 166 2 32 92 161 120 97 49 108 60
116 119 168 188 69 104 4
```

2-regret - cycle

min=111748 max=129972 avg=120502.

best = 111748



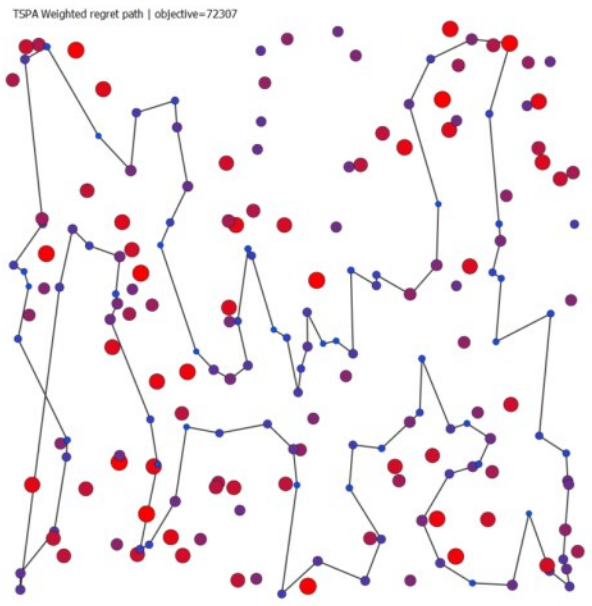
Best solution nodes:

```
171 187 30 23 162 90 123 152 108 11 97 120
110 129 168 116 82 77 199 48 126 137 80 38
91 157 4 196 51 133 139 112 5 115 175 39
154 62 34 109 186 136 197 163 19 29 17 55
86 155 21 147 47 159 130 117 93 50 1 128
164 194 174 81 43 113 89 124 107 75 26 78
88 64 57 105 182 138 111 192 58 135 73 41
99 177 52 144 45 185 102 87 96 98 114 22
10 183 166 2 171
```

Weighted regret - path

min=72307 max=78176 avg=75189.33

best = 72307



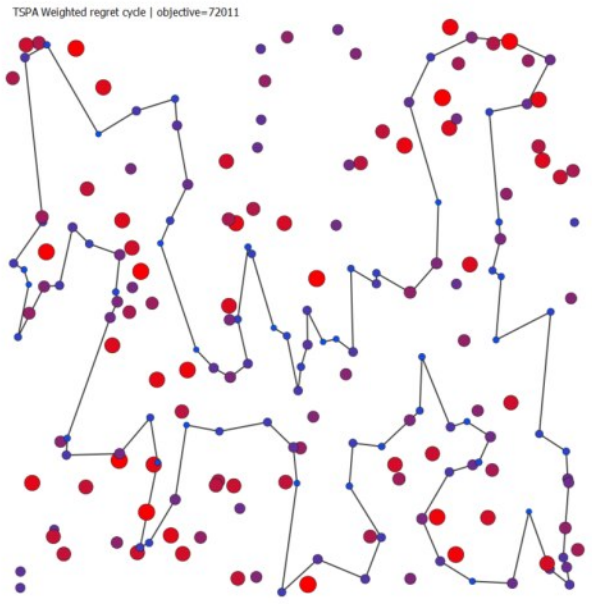
Best solution nodes:

```
49 108 152 100 123 83 160 30 187 171 2 189
141 87 45 73 135 114 181 192 158 58 84 134
72 176 174 150 106 113 89 42 151 138 182
57 64 107 88 78 193 153 75 170 146 159 130
47 93 50 3 169 1 178 29 163 136 186 164
142 8 9 54 33 131 36 177 52 179 53 140 144
67 16 51 35 185 102 28 180 132 76 4 71 157
91 38 80 48 31 60 116 149 63 168 97 127 69
129 104 49
```

Weighted regret - cycle

min=72011 max=76027 avg=73335.27

best = 72011



Best solution nodes:

```
60 116 149 63 168 11 49 108 152 100 123 83
160 97 127 161 30 187 171 2 189 141 87 45
73 135 114 181 192 158 58 84 134 72 176
174 150 106 113 89 42 151 138 182 57 64
107 88 78 193 153 75 170 146 159 130 47 93
50 3 169 1 178 101 55 29 163 136 186 164
142 8 9 54 33 131 36 177 52 179 53 140 144
67 16 51 35 185 102 28 180 132 76 4 71 157
91 80 48 31 60
```

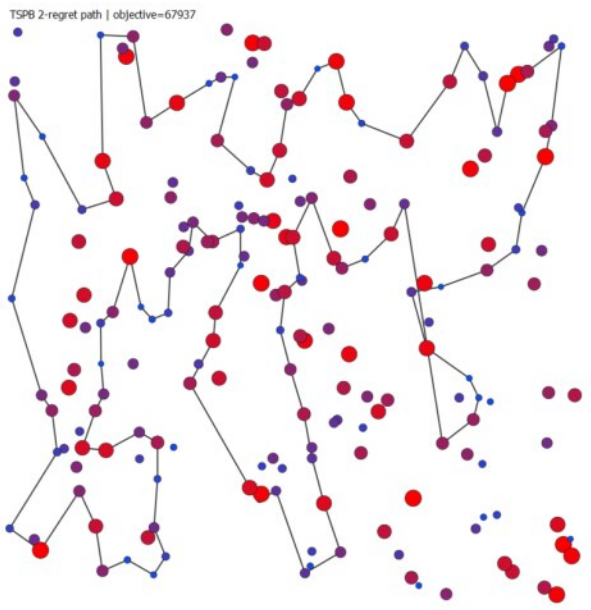
4. Best solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

2-regret - path

min=67937 max=79102 avg=73139.61

best = 67937



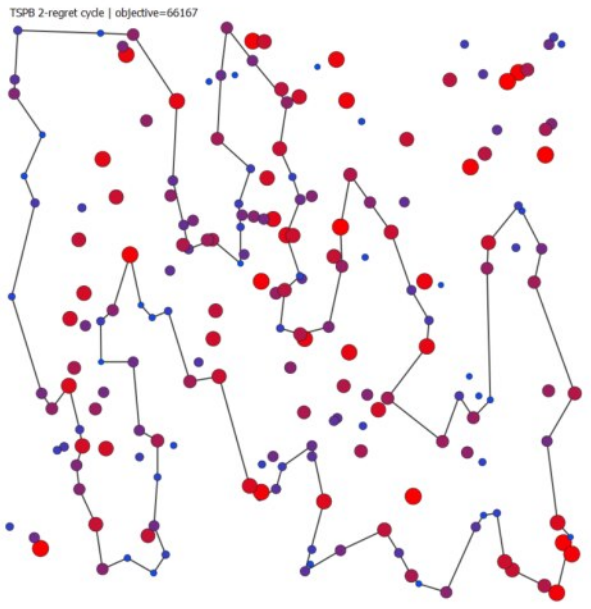
Best solution nodes:

```
150 52 129 127 106 162 54 78 10 114 193 61
105 169 165 199 187 70 190 194 77 57 83 44
89 32 156 166 92 120 63 96 198 121 195 50
42 116 123 15 158 149 13 28 145 65 178 103
132 1 108 36 91 62 175 134 45 157 60 141
155 113 5 180 115 170 125 26 161 154 140
47 38 21 183 168 172 69 124 4 33 107 85 51
82 88 126 181 25 98 151 173 184 137 86 75
135 76 144 104 150
```

2-regret - cycle

min=66167 max=77274 avg=72327.22

best = 66167



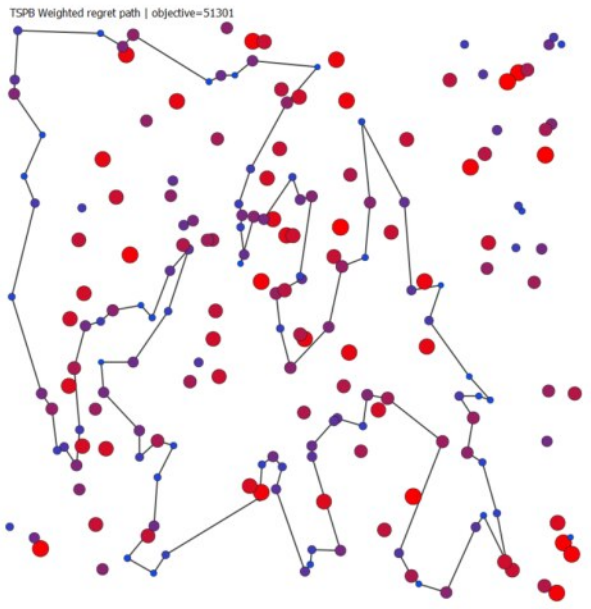
Best solution nodes:

```
41 196 114 148 143 131 118 68 110 23 93 9
188 14 20 74 17 171 39 99 163 88 82 126
152 167 51 85 107 49 33 47 140 154 161 26
125 170 97 155 141 60 157 45 134 175 62 91
36 197 5 6 102 103 178 65 145 28 149 13 37
116 42 195 130 30 21 168 69 172 133 120 63
198 58 48 156 166 72 46 109 184 173 151
111 27 135 0 79 185 144 162 192 106 84 150
80 52 90 78 41
```

Weighted regret - path

min=51301 max=65594 avg=57804.59

best = 51301



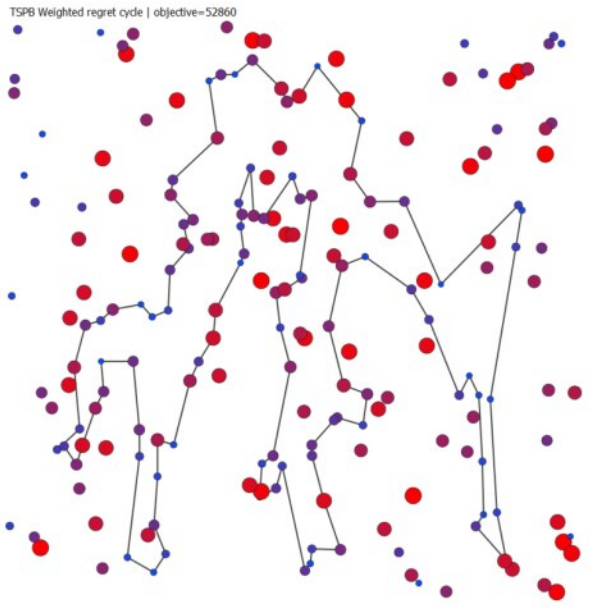
Best solution nodes:

```
121 198 96 48 89 156 120 133 136 172 160
69 8 117 72 46 86 184 3 122 151 98 27 135
76 185 57 104 162 54 192 127 90 129 80 52
182 153 14 2 20 74 17 171 39 99 150 84 66
67 24 34 152 181 88 179 87 82 51 167 73 59
85 45 134 175 157 60 40 141 71 155 115 170
97 47 21 38 140 154 26 125 177 19 6 197 29
132 103 178 65 145 28 149 13 146 37 116
147 42 121
```

Weighted regret - cycle

min=52860 max=63406 avg=57903.04

best = 52860



Best solution nodes:

```
5 197 29 132 6 19 177 125 26 154 140 47 38
21 30 183 191 130 63 121 198 96 48 43 156
32 89 57 79 185 104 54 196 114 10 90 14 2
74 20 153 129 127 80 192 162 76 135 27 11
66 67 24 34 152 181 126 88 179 87 82 167
51 85 59 73 98 151 122 173 3 184 86 46 72
117 8 120 133 136 172 160 69 124 4 64 33
40 141 60 157 45 134 175 71 155 97 170 115
180 5
```

5. Conclusions

The experimental results show that regret-based selection can improve the greedy baselines when combined with the best objective change.
The effect depends on whether the method constructs an open path or a closed cycle.

Main conclusions

Weighted regret cycle achieved the best individual solution on TSPA (72011), while weighted regret path achieved the best individual solution on TSPB (51301).

The best average results were also obtained by the weighted methods: weighted regret cycle on TSPA (73335.27) and weighted regret path on TSPB (57804.59).

Pure 2-regret performs worse on these instances because it prioritizes nodes with a large difference between their first and second insertion choices, even when their best insertion is expensive. The weighted criterion restores pressure toward a low immediate objective change.

The open-path and closed-cycle variants behave differently. The cycle variant accounts for every insertion edge in the current closed route, while the path variant follows the extended nearest-neighbor construction used in lab 01.

All 3,200 generated routes passed the internal structural validation. The saved routes and plots provide a reproducible starting point for local search and later evolutionary methods.

SOURCE REPOSITORY

https://github.com/Flat-Earther/Evolutionary-Computation/tree/main/lab-02-greedy-regret-heuristics

The large-format best and worst visualizations are presented in the final appendix pages.

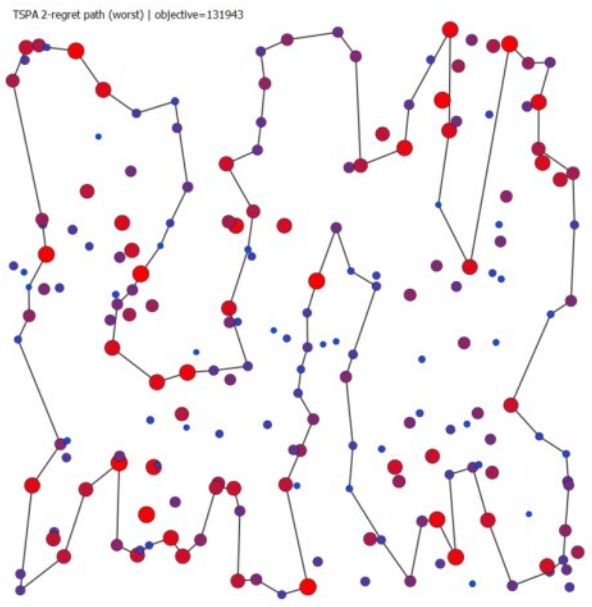
6. Worst solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

2-regret - path

min=112968 max=131943 avg=121241.

worst = 131943



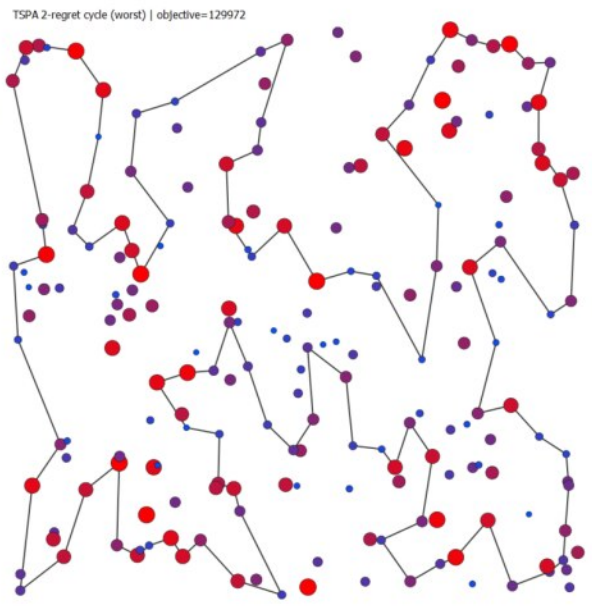
Worst solution nodes:

```
128 164 56 197 186 109 15 154 39 175 115 5
112 139 18 133 51 172 185 28 191 23 12 83
90 68 132 76 4 157 91 137 126 48 199 77 82
167 63 119 168 95 188 104 110 120 125 92
32 2 166 183 10 96 70 98 22 25 114 118 121
61 41 140 53 179 52 74 143 33 54 131 99 84
58 158 111 138 165 105 151 42 124 64 88
153 75 170 159 130 117 47 37 147 21 46 86
55 17 79 128
```

2-regret - cycle

min=111748 max=129972 avg=120502.

worst = 129972



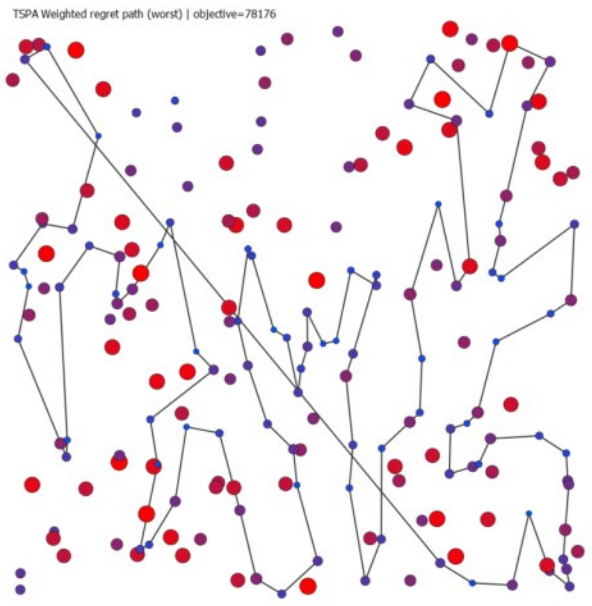
Worst solution nodes:

```
175 112 139 18 196 16 190 74 33 9 174 142
164 34 186 136 197 163 19 79 17 55 86 46 6
147 37 47 169 128 93 43 117 130 159 170 75
153 88 64 124 105 182 111 158 138 0 72 13
134 84 99 179 41 73 45 185 156 28 191 23
141 87 27 70 98 114 22 10 183 166 2 32 92
125 120 110 104 188 95 168 116 167 82 77
199 7 126 137 80 66 108 152 14 68 76 38 91
157 115 175
```

Weighted regret - path

min=72307 max=78176 avg=75189.33

worst = 78176



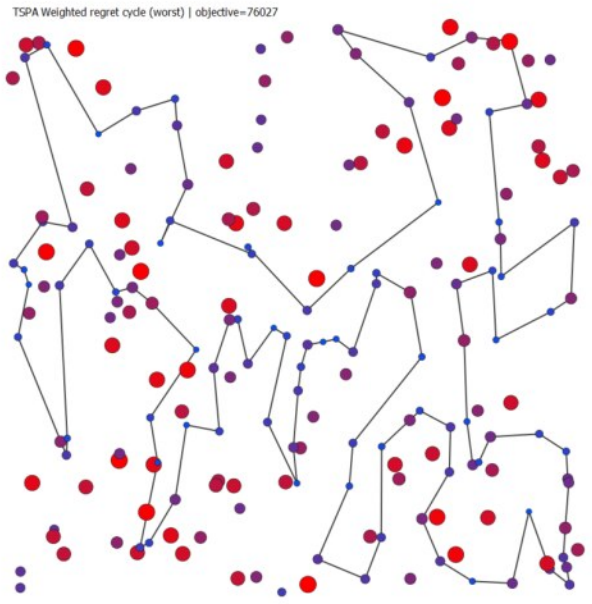
Worst solution nodes:

```
182 57 64 107 88 78 193 153 75 170 146 159
130 113 89 42 151 150 106 43 93 47 37 147
50 3 169 1 59 101 55 29 178 136 186 184
128 85 164 8 174 176 72 134 158 192 58 84
99 131 9 54 33 36 177 52 179 53 140 144 67
16 51 35 185 45 73 135 181 114 25 98 87
141 189 2 171 187 30 28 180 76 132 90 83
123 100 152 49 97 127 168 63 149 116 60
108 80 48 31 182
```

Weighted regret - cycle

min=72011 max=76027 avg=73335.27

worst = 76027



Worst solution nodes:

```
154 186 164 33 52 51 16 76 132 4 71 157 91
80 48 31 108 60 116 149 63 168 127 97 49
152 123 90 162 180 30 187 171 2 189 141 87
28 156 35 185 67 144 45 135 73 140 53 179
177 36 131 54 9 8 174 84 58 181 192 158
134 72 176 150 151 138 182 57 64 107 88 78
193 153 75 170 146 159 130 113 89 42 106
81 85 3 93 47 37 147 50 169 1 178 101 29
163 136 39 154
```

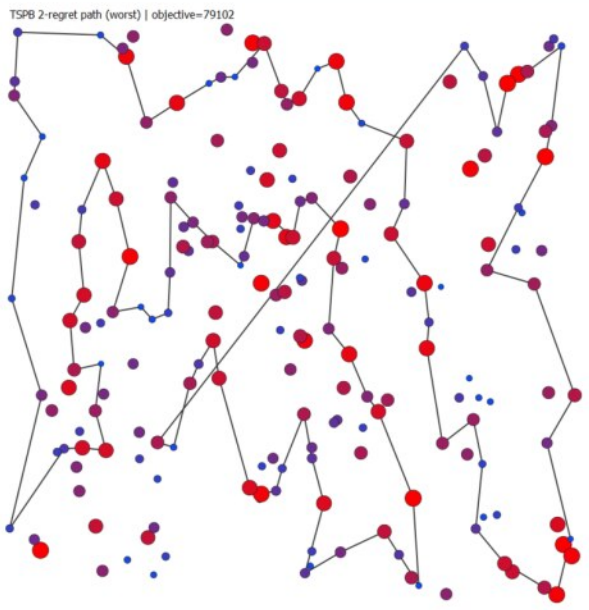
6. Worst solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

2-regret - path

min=67937 max=79102 avg=73139.61

worst = 79102



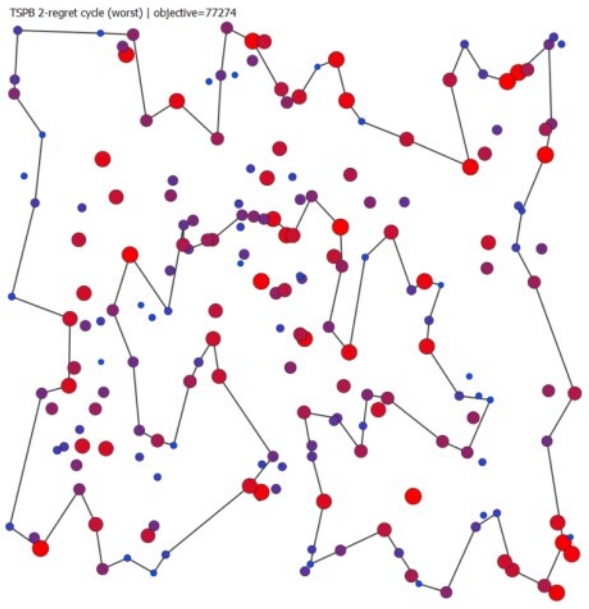
Worst solution nodes:

```
141 40 33 64 4 49 107 101 51 167 25 181
126 82 88 163 171 186 119 66 35 27 75 0 86
46 137 95 8 69 168 183 191 38 47 140 154
26 161 15 123 158 138 55 53 19 170 180 113
5 29 1 178 65 28 149 13 37 116 7 50 195
121 198 96 164 156 32 89 44 83 57 77 104
144 159 192 106 150 52 153 74 188 93 23
110 118 131 143 78 196 193 61 105 169 165
199 187 70 190 141
```

2-regret - cycle

min=66167 max=77274 avg=72327.22

worst = 77274



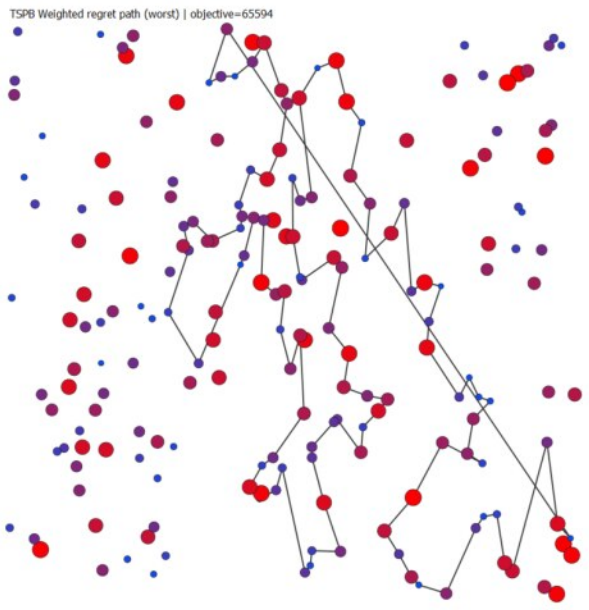
Worst solution nodes:

```
176 30 47 161 26 97 155 141 40 33 64 4 49
73 107 45 134 175 62 91 36 108 1 178 102
53 65 145 149 13 37 116 42 50 195 63 198
58 16 156 32 89 44 83 57 77 12 194 190 70
199 165 81 61 193 114 10 143 131 118 100
93 9 14 74 17 171 99 163 88 87 126 181 25
24 31 66 84 150 182 90 80 106 192 54 159
162 144 76 35 27 135 0 86 137 95 8 136 168
21 176
```

Weighted regret - path

min=51301 max=65594 avg=57804.59

worst = 65594



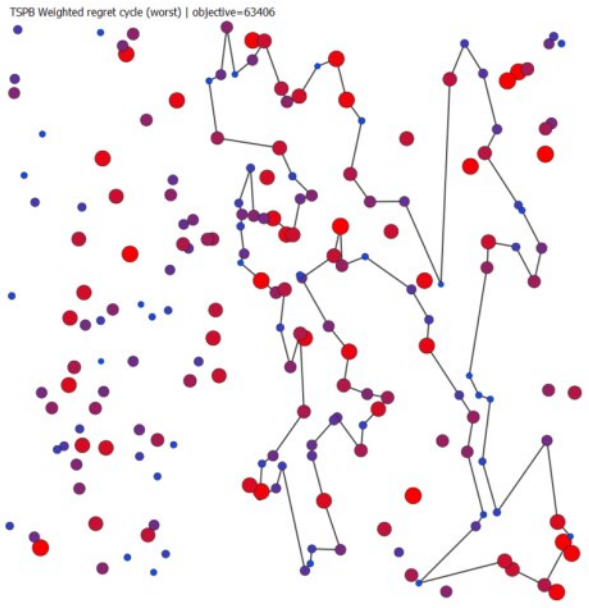
Worst solution nodes:

```
58 121 198 96 48 164 43 156 166 92 120 133
136 172 168 183 30 21 47 64 69 160 8 117
128 122 173 151 98 111 25 73 59 107 85 51
167 82 87 179 88 126 181 152 34 24 31 67
119 84 66 11 35 27 135 75 3 184 137 72 46
86 32 89 44 83 57 79 185 76 144 104 162
159 54 192 106 80 127 129 90 52 182 153
150 186 163 99 39 171 17 74 20 14 188 2 9
118 68 110 58
```

Weighted regret - cycle

min=52860 max=63406 avg=57903.04

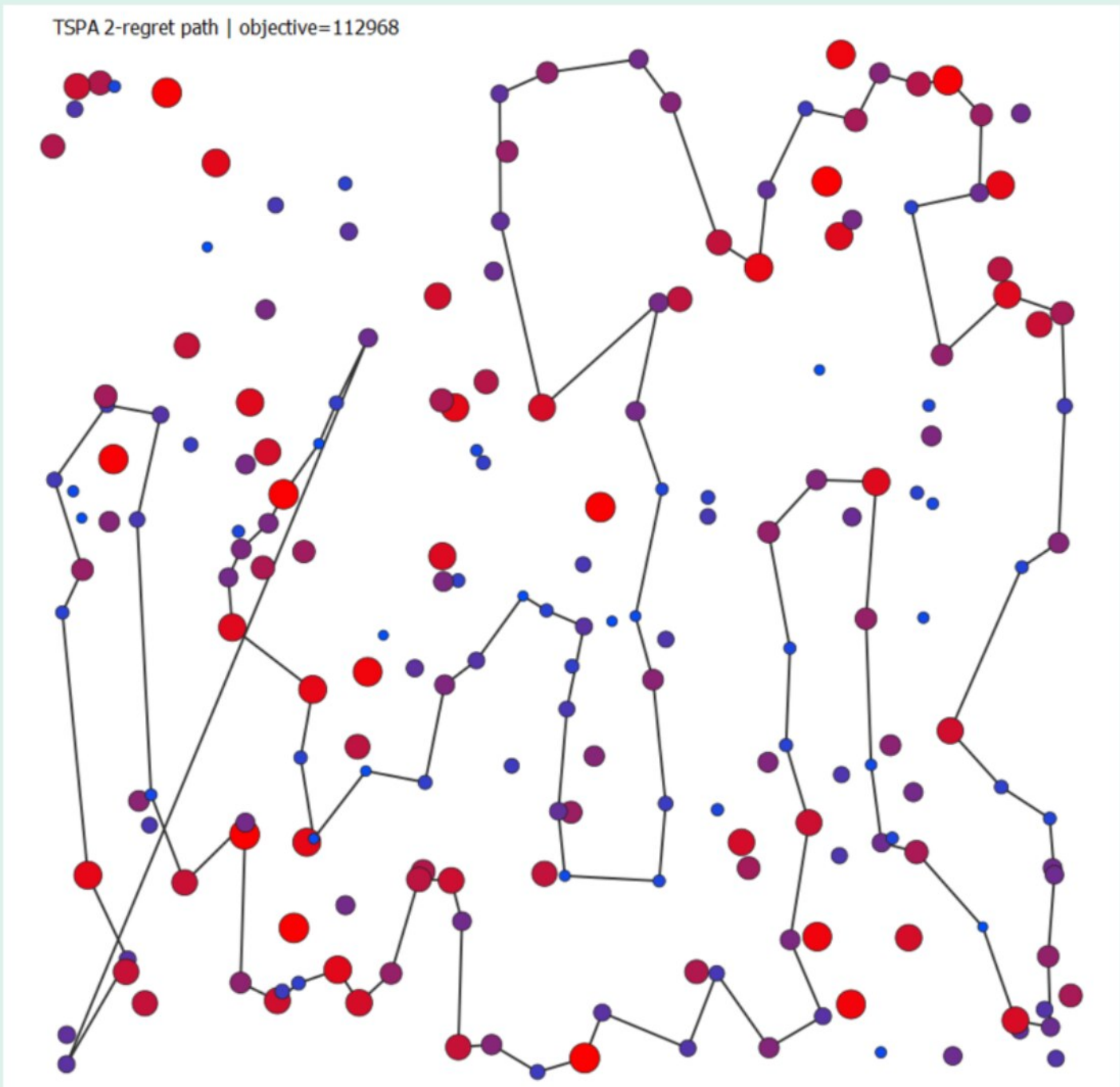
worst = 63406



Worst solution nodes:

```
0 135 76 162 192 106 80 52 182 20 74 171
188 2 9 93 110 68 118 14 153 90 129 127 78
41 10 143 148 114 196 18 187 70 190 194 54
104 185 79 57 83 44 89 32 156 43 164 48 96
58 198 121 63 166 72 86 46 137 95 117 8
120 133 136 172 160 69 128 122 173 151 98
111 25 73 59 85 51 167 82 87 179 88 126
181 152 34 24 31 67 119 84 66 11 35 27 184
3 75 0
```

7. Large best visualizations - TSPA



BEST SOLUTION

2-regret - path

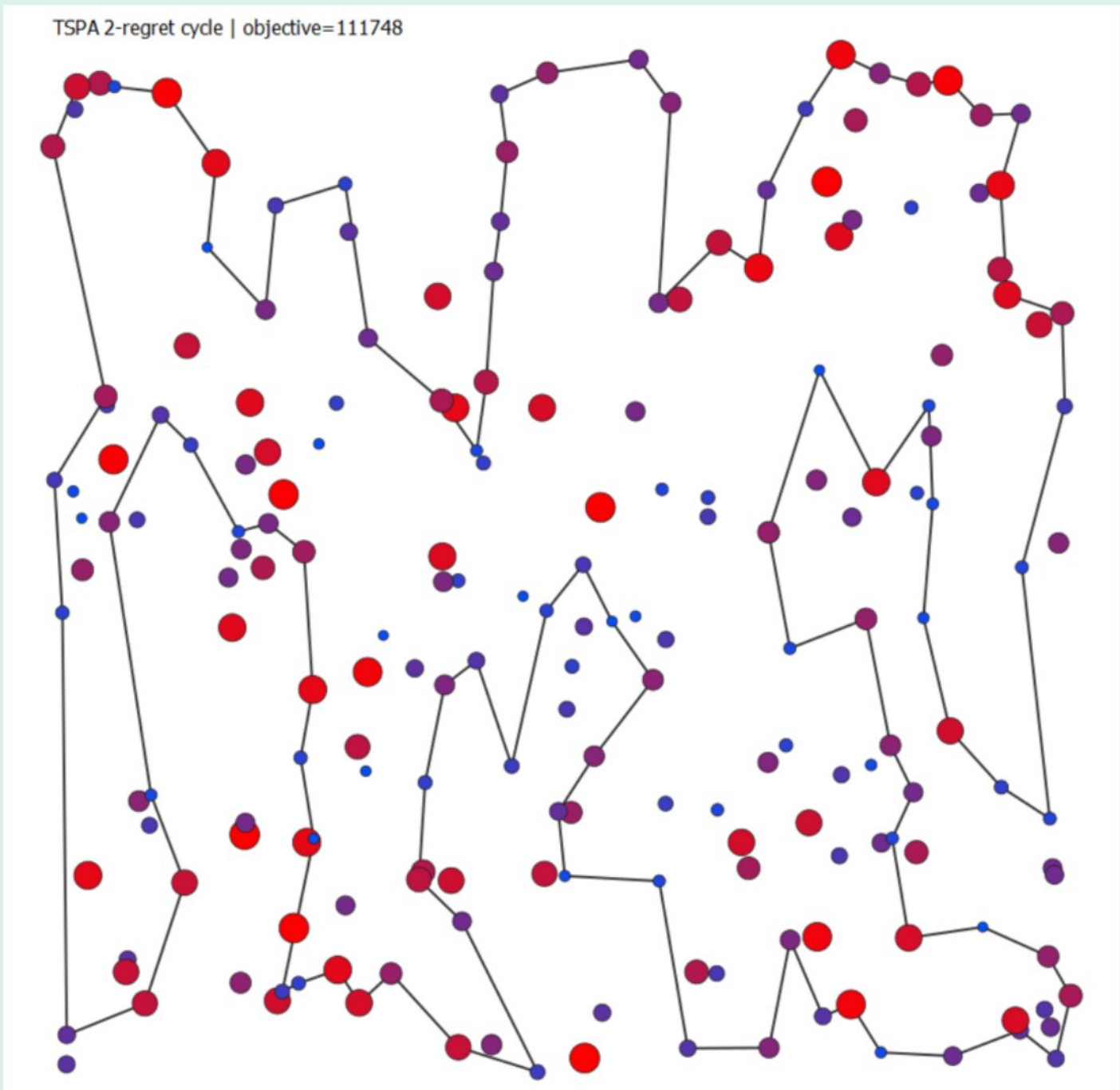
min=112968
max=131943
avg=121241.39

objective 112968

Best solution nodes:

4 76 132 68 90 83 160 12
23 30 187 141 87 102 185
67 144 179 140 73 135 58
84 99 36 33 143 62 190
112 115 175 39 154 34 109
186 136 24 163 19 29 17
101 178 59 155 21 147 37
47 117 130 159 170 75 193
122 107 103 42 106 81 128
142 194 174 176 0 138 182
111 158 192 181 118 114
25 22 98 70 96 10 183 166
2 32 92 161 120 97 49 108
60 116 119 168 188 69 104
4

100 selected nodes; closed route



BEST SOLUTION

2-regret - cycle

min=111748
max=129972
avg=120502.06

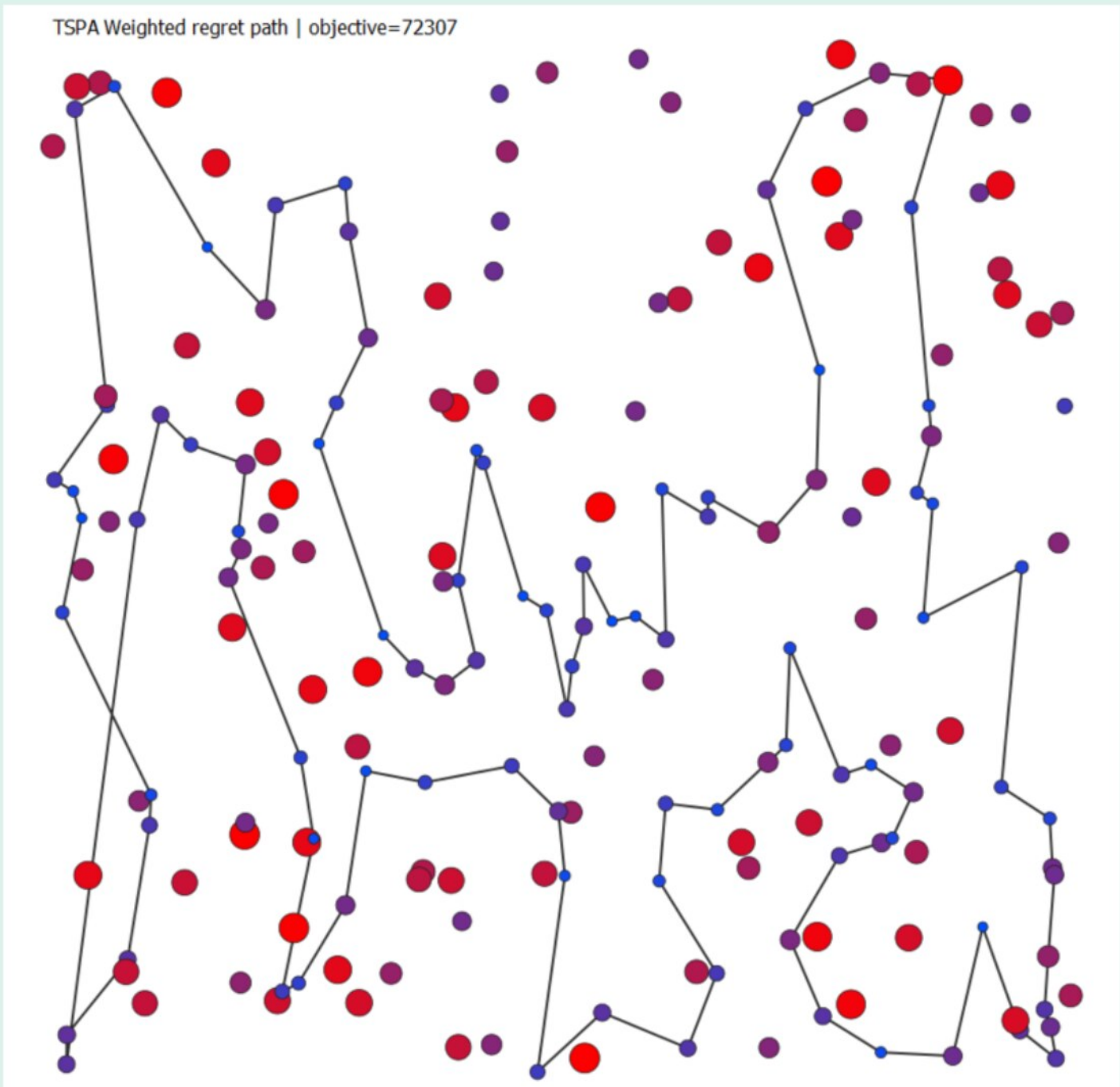
objective 111748

Best solution nodes:

171 187 30 23 162 90 123
152 108 11 97 120 110 129
168 116 82 77 199 48 126
137 80 38 91 157 4 196 51
133 139 112 5 115 175 39
154 62 34 109 186 136 197
163 19 29 17 55 86 155 21
147 47 159 130 117 93 50
1 128 164 194 174 81 43
113 89 124 107 75 26 78
88 64 57 105 182 138 111
192 58 135 73 41 99 177
52 144 45 185 102 87 96
98 114 22 10 183 166 2
171

100 selected nodes; closed route

7. Large best visualizations - TSPA



BEST SOLUTION

Weighted regret - path

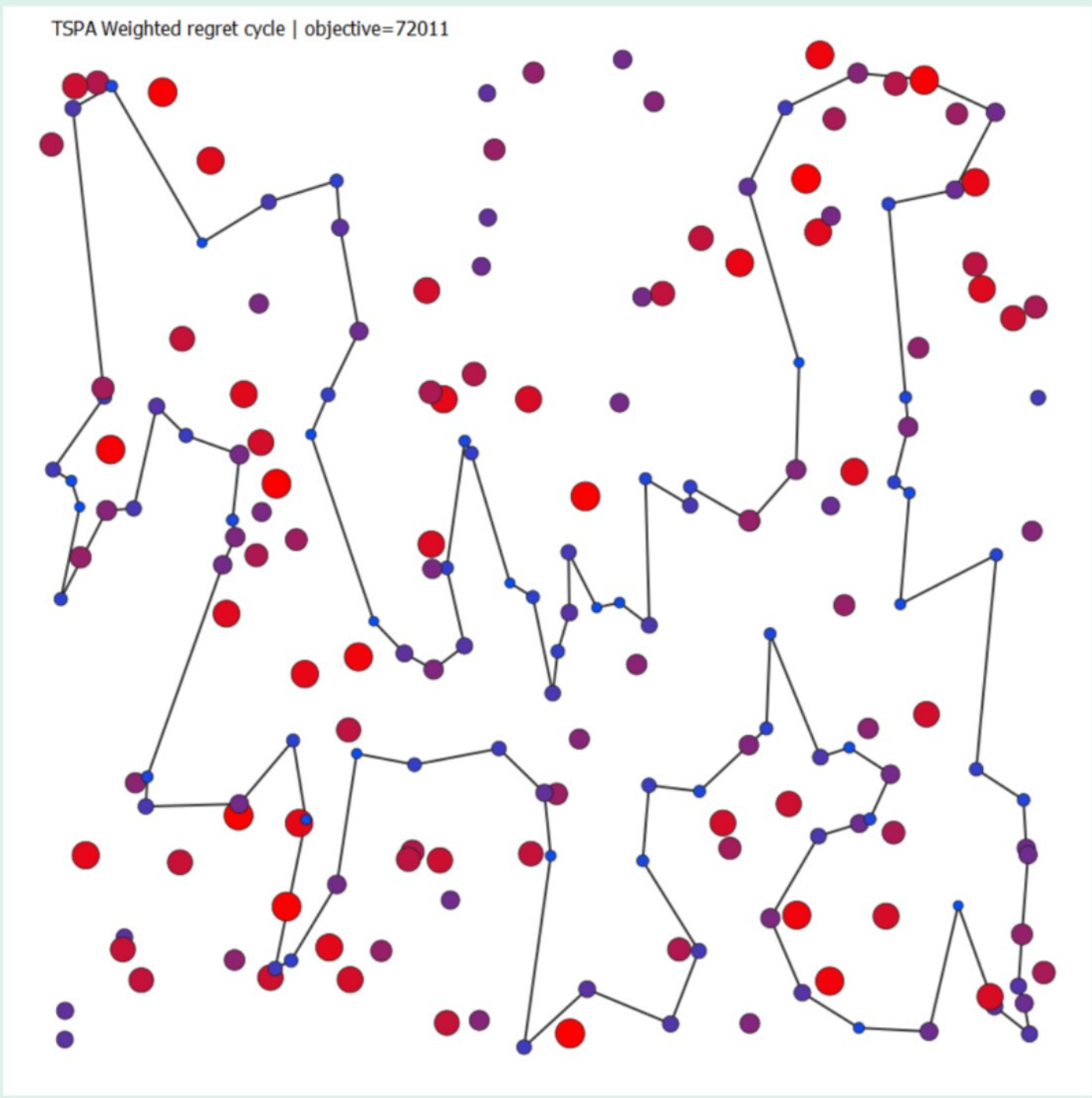
min=72307
max=78176
avg=75189.33

objective 72307

Best solution nodes:

```
49 108 152 100 123 83 160
30 187 171 2 189 141 87
45 73 135 114 181 192 158
58 84 134 72 176 174 150
106 113 89 42 151 138 182
57 64 107 88 78 193 153
75 170 146 159 130 47 93
50 3 169 1 178 29 163 136
186 164 142 8 9 54 33 131
36 177 52 179 53 140 144
67 16 51 35 185 102 28
180 132 76 4 71 157 91 38
80 48 31 60 116 149 63
168 97 127 69 129 104 49
```

100 selected nodes; closed route



BEST SOLUTION

Weighted regret - cycle

min=72011
max=76027
avg=73335.27

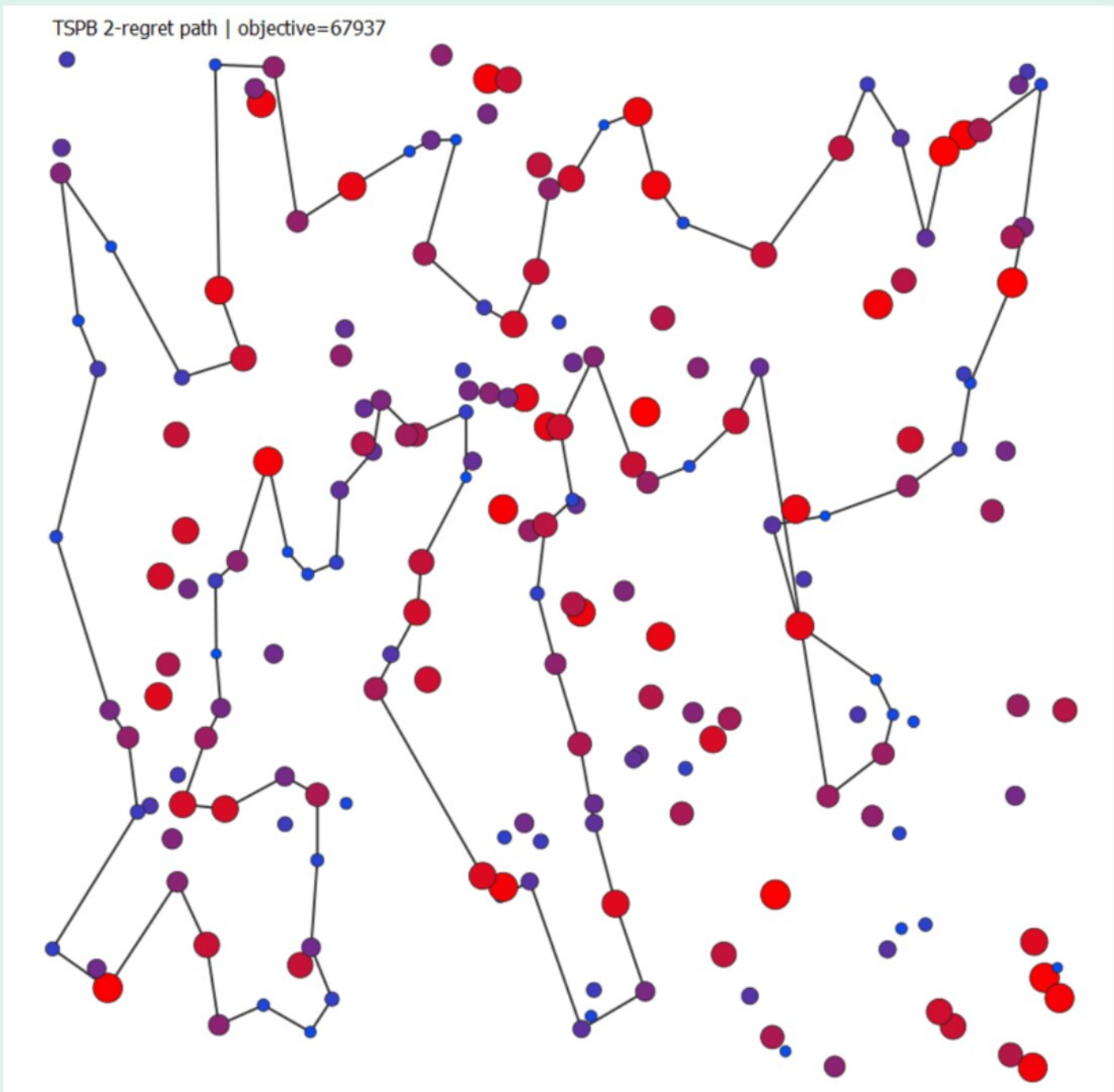
objective 72011

Best solution nodes:

```
60 116 149 63 168 11 49
108 152 100 123 83 160 97
127 161 30 187 171 2 189
141 87 45 73 135 114 181
192 158 58 84 134 72 176
174 150 106 113 89 42 151
138 182 57 64 107 88 78
193 153 75 170 146 159
130 47 93 50 3 169 1 178
101 55 29 163 136 186 164
142 8 9 54 33 131 36 177
52 179 53 140 144 67 16
51 35 185 102 28 180 132
76 4 71 157 91 80 48 31
60
```

100 selected nodes; closed route

7. Large best visualizations - TSPB



BEST SOLUTION

2-regret - path

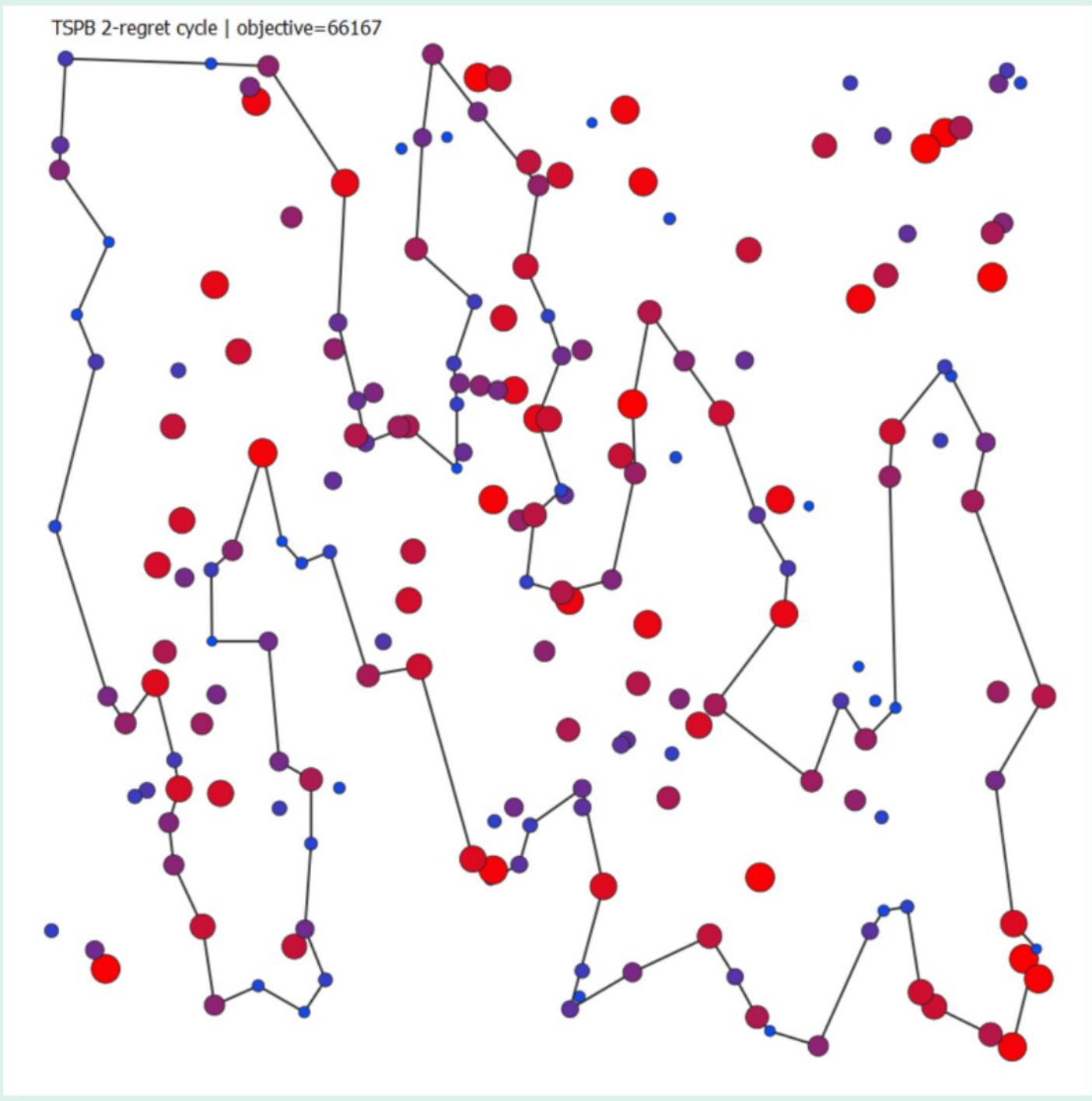
min=67937
max=79102
avg=73139.61

objective 67937

Best solution nodes:

```
150 52 129 127 106 162 54
78 10 114 193 61 105 169
165 199 187 70 190 194 77
57 83 44 89 32 156 166 92
120 63 96 198 121 195 50
42 116 123 15 158 149 13
28 145 65 178 103 132 1
108 36 91 62 175 134 45
157 60 141 155 113 5 180
115 170 125 26 161 154
140 47 38 21 183 168 172
69 124 4 33 107 85 51 82
88 126 181 25 98 151 173
184 137 86 75 135 76 144
104 150
```

100 selected nodes; closed route



BEST SOLUTION

2-regret - cycle

min=66167
max=77274
avg=72327.22

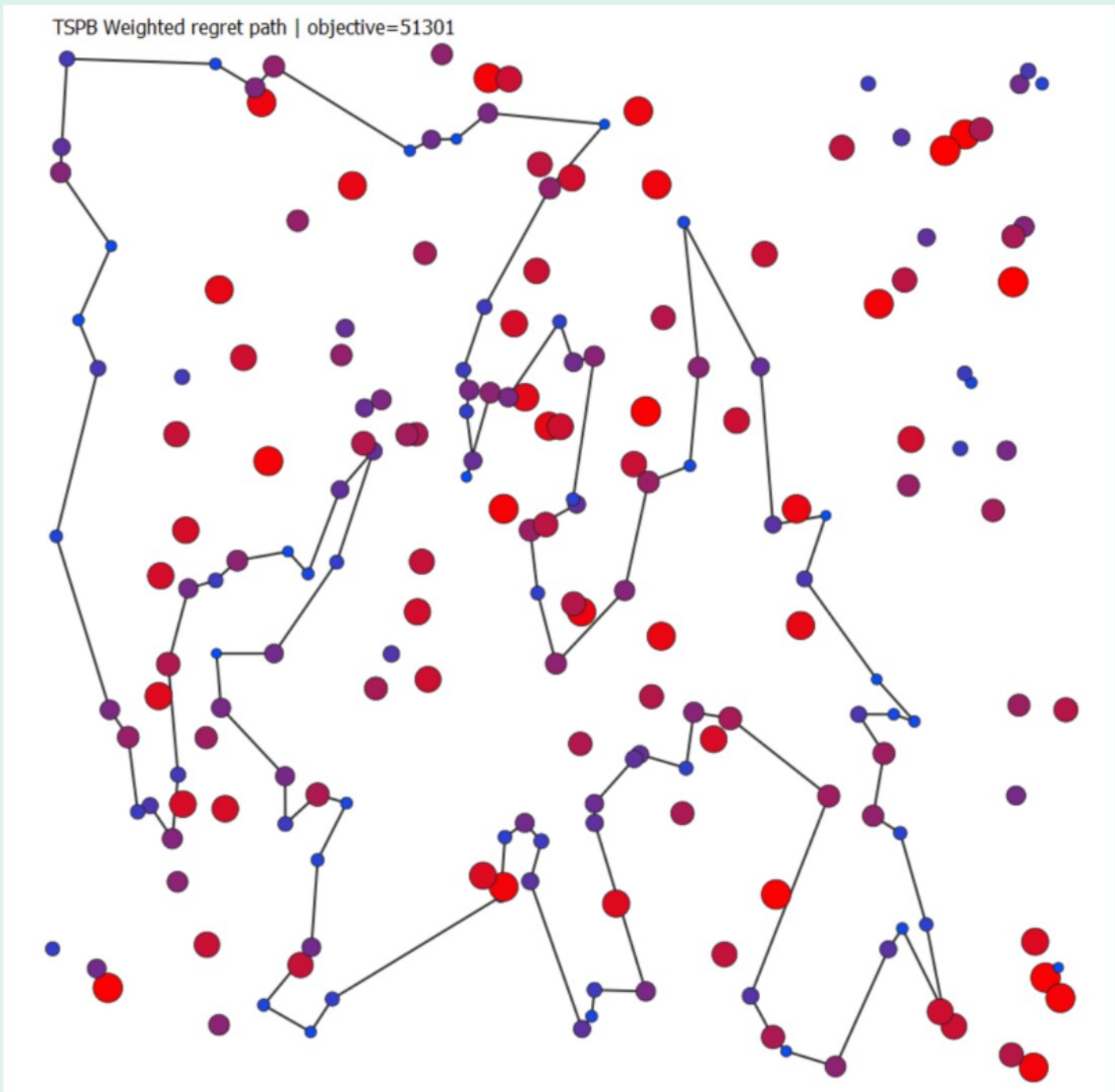
objective 66167

Best solution nodes:

```
41 196 114 148 143 131
118 68 110 23 93 9 188 14
20 74 17 171 39 99 163 88
82 126 152 167 51 85 107
49 33 47 140 154 161 26
125 170 97 155 141 60 157
45 134 175 62 91 36 197 5
6 102 103 178 65 145 28
149 13 37 116 42 195 130
30 21 168 69 172 133 120
63 198 58 48 156 166 72
46 109 184 173 151 111 27
135 0 79 185 144 162 192
106 84 150 80 52 90 78 41
```

100 selected nodes; closed route

7. Large best visualizations - TSPB



BEST SOLUTION

Weighted regret - path

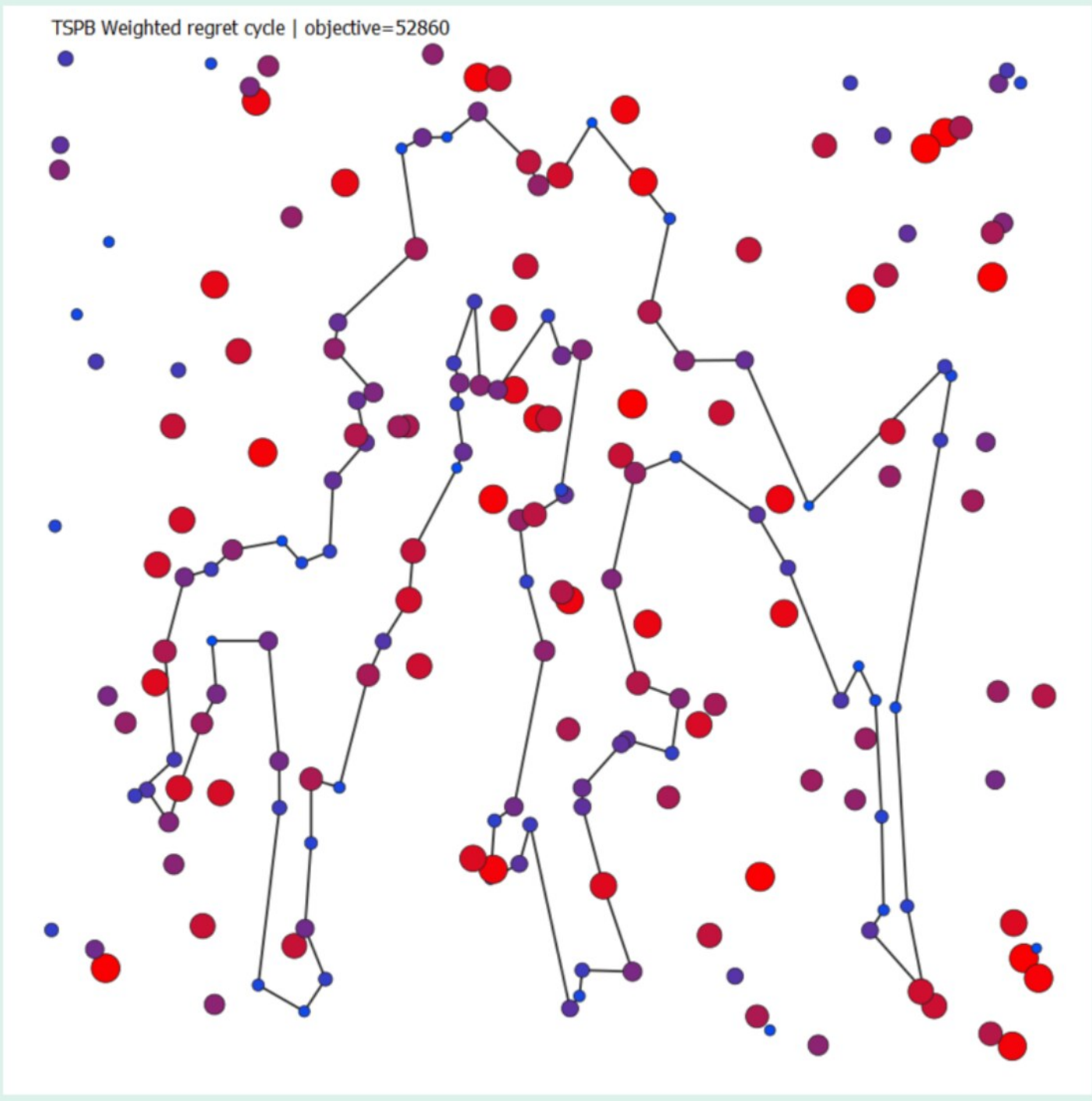
min=51301
max=65594
avg=57804.59

objective 51301

Best solution nodes:

```
121 198 96 48 89 156 120
133 136 172 160 69 8 117
72 46 86 184 3 122 151 98
27 135 76 185 57 104 162
54 192 127 90 129 80 52
182 153 14 2 20 74 17 171
39 99 150 84 66 67 24 34
152 181 88 179 87 82 51
167 73 59 85 45 134 175
157 60 40 141 71 155 115
170 97 47 21 38 140 154
26 125 177 19 6 197 29
132 103 178 65 145 28 149
13 146 37 116 147 42 121
```

100 selected nodes; closed route



BEST SOLUTION

Weighted regret - cycle

min=52860
max=63406
avg=57903.04

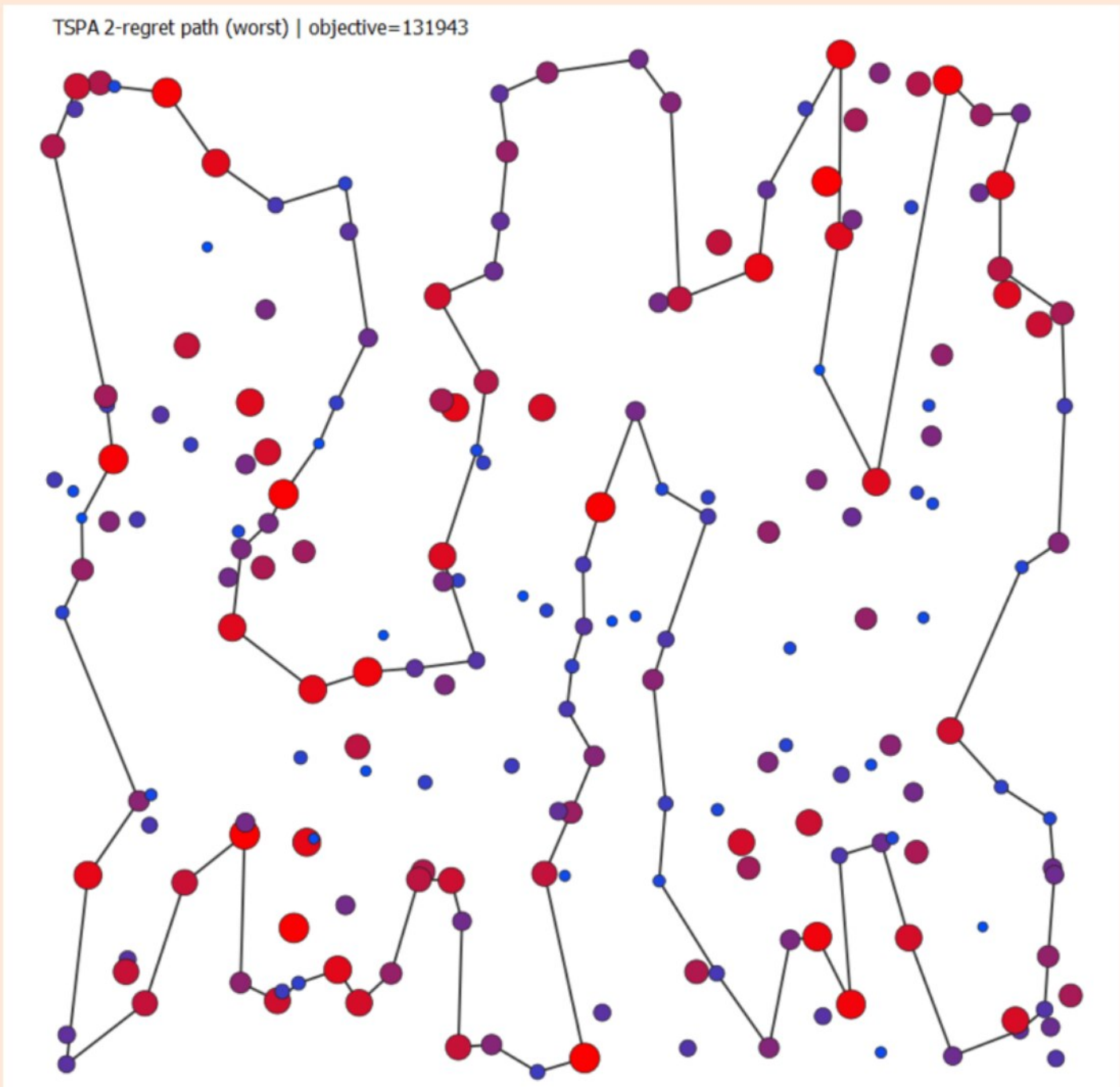
objective 52860

Best solution nodes:

```
5 197 29 132 6 19 177 125
26 154 140 47 38 21 30
183 191 130 63 121 198 96
48 43 156 32 89 57 79 185
104 54 196 114 10 90 14 2
74 20 153 129 127 80 192
162 76 135 27 11 66 67 24
34 152 181 126 88 179 87
82 167 51 85 59 73 98 151
122 173 3 184 86 46 72
117 8 120 133 136 172 160
69 124 4 64 33 40 141 60
157 45 134 175 71 155 97
170 115 180 5
```

100 selected nodes; closed route

7. Large worst visualizations - TSPA



WORST SOLUTION

2-regret - path

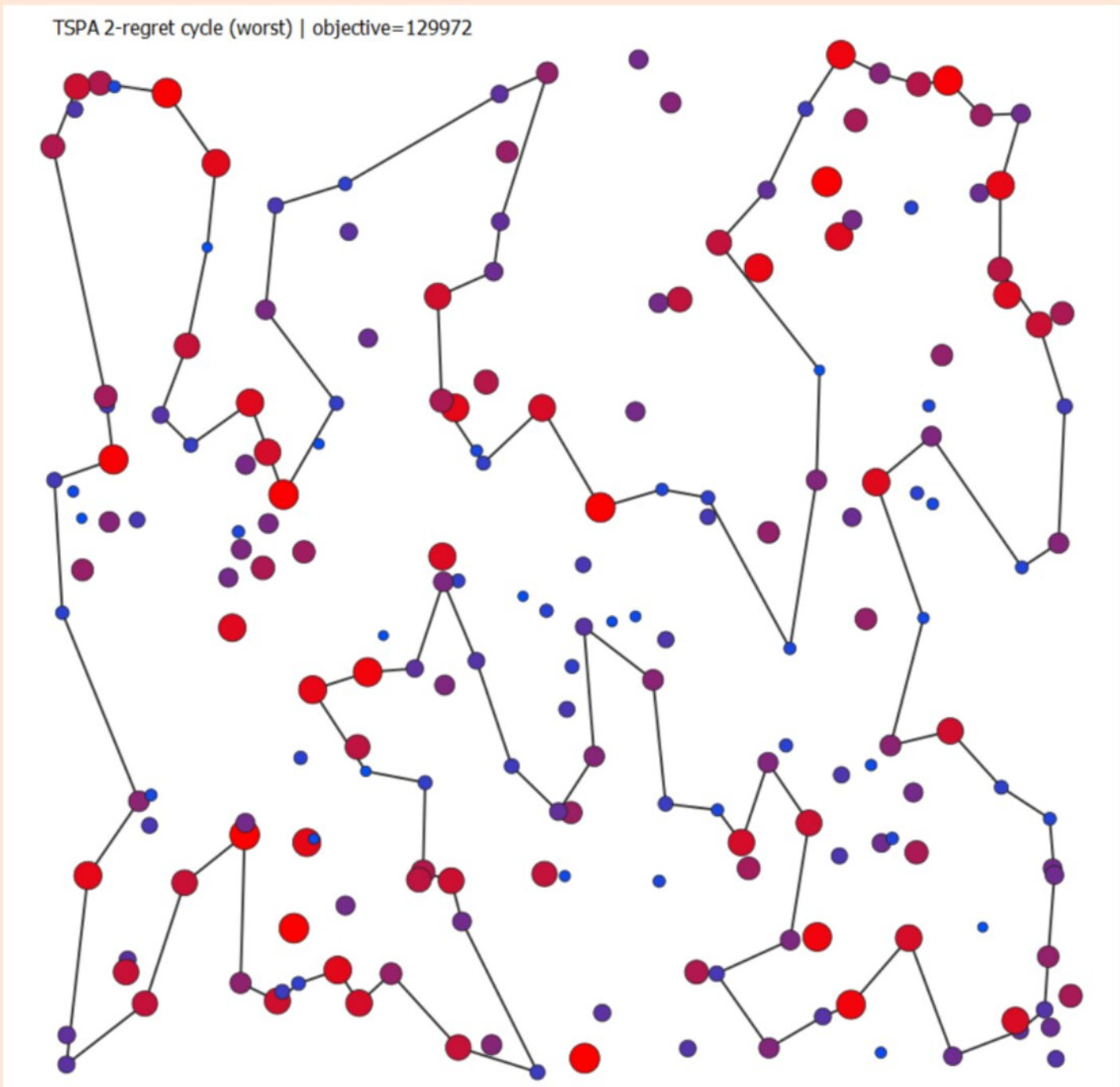
min=112968
max=131943
avg=121241.39

objective 131943

Worst solution nodes:

128 164 56 197 186 109 15
154 39 175 115 5 112 139
18 133 51 172 185 28 191
23 12 83 90 68 132 76 4
157 91 137 126 48 199 77
82 167 63 119 168 95 188
104 110 120 125 92 32 2
166 183 10 96 70 98 22 25
114 118 121 61 41 140 53
179 52 74 143 33 54 131
99 84 58 158 111 138 165
105 151 42 124 64 88 153
75 170 159 130 117 47 37
147 21 46 86 55 17 79 128

100 selected nodes; closed route



WORST SOLUTION

2-regret - cycle

min=111748
max=129972
avg=120502.06

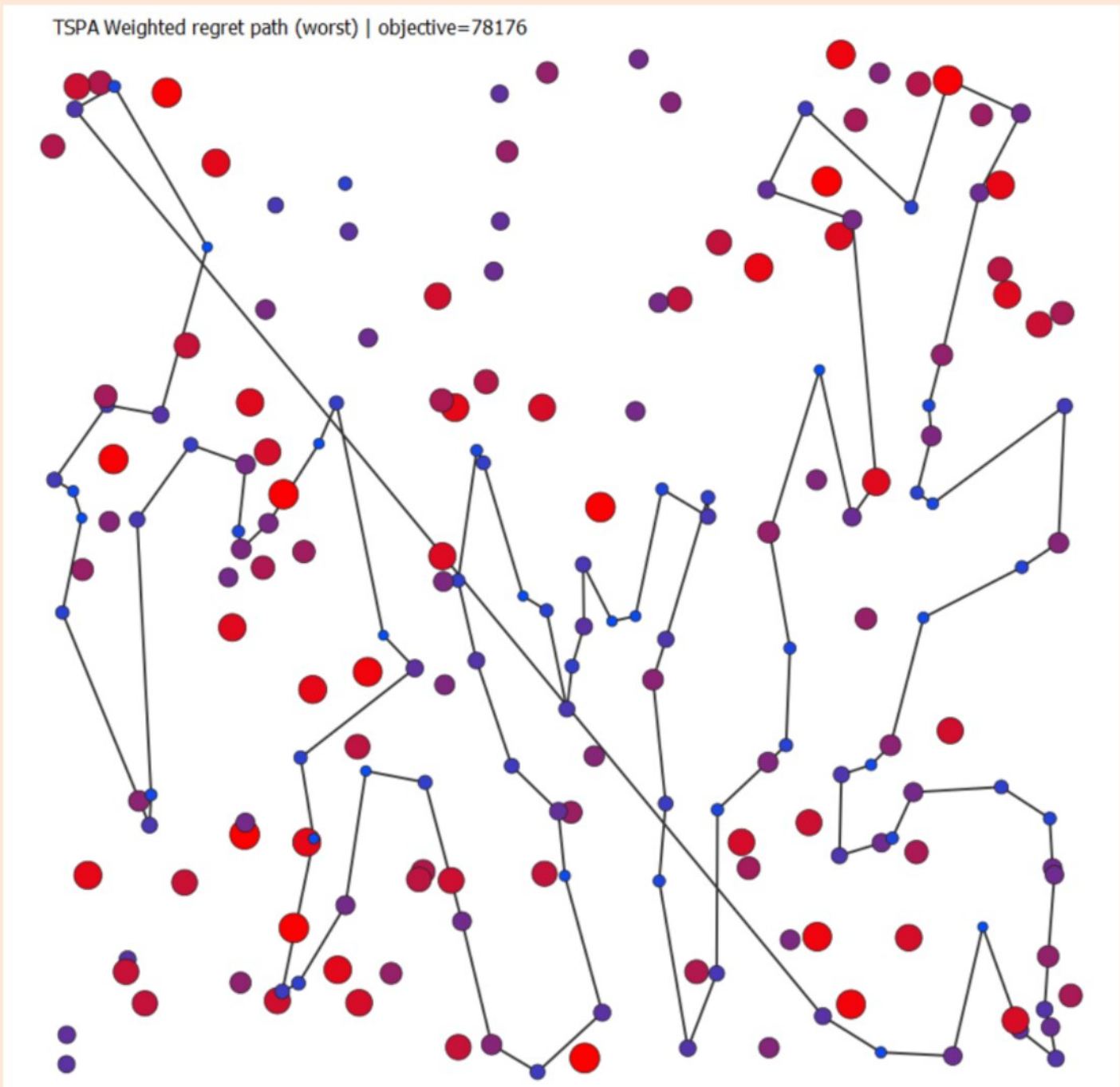
objective 129972

Worst solution nodes:

175 112 139 18 196 16 190
74 33 9 174 142 164 34
186 136 197 163 19 79 17
55 86 46 6 147 37 47 169
128 93 43 117 130 159 170
75 153 88 64 124 105 182
111 158 138 0 72 13 134
84 99 179 41 73 45 185
156 28 191 23 141 87 27
70 98 114 22 10 183 166 2
32 92 125 120 110 104 188
95 168 116 167 82 77 199
7 126 137 80 66 108 152
14 68 76 38 91 157 115
175

100 selected nodes; closed route

7. Large worst visualizations - TSPA



WORST SOLUTION

Weighted regret - path

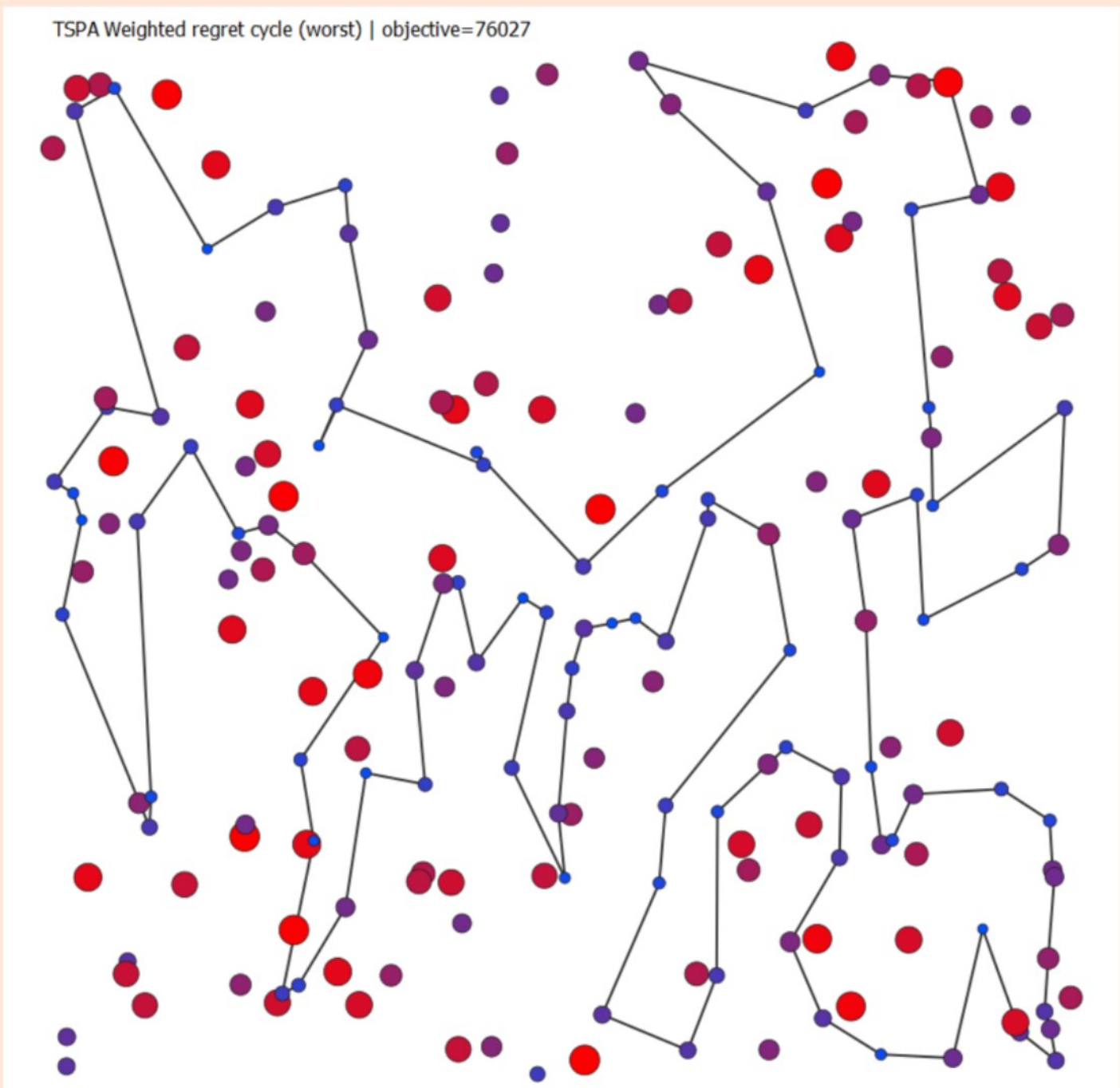
min=72307
max=78176
avg=75189.33

objective 78176

Worst solution nodes:

182 57 64 107 88 78 193
153 75 170 146 159 130
113 89 42 151 150 106 43
93 47 37 147 50 3 169 1
59 101 55 29 178 136 186
184 128 85 164 8 174 176
72 134 158 192 58 84 99
131 9 54 33 36 177 52 179
53 140 144 67 16 51 35
185 45 73 135 181 114 25
98 87 141 189 2 171 187
30 28 180 76 132 90 83
123 100 152 49 97 127 168
63 149 116 60 108 80 48
31 182

100 selected nodes; closed route



WORST SOLUTION

Weighted regret - cycle

min=72011
max=76027
avg=73335.27

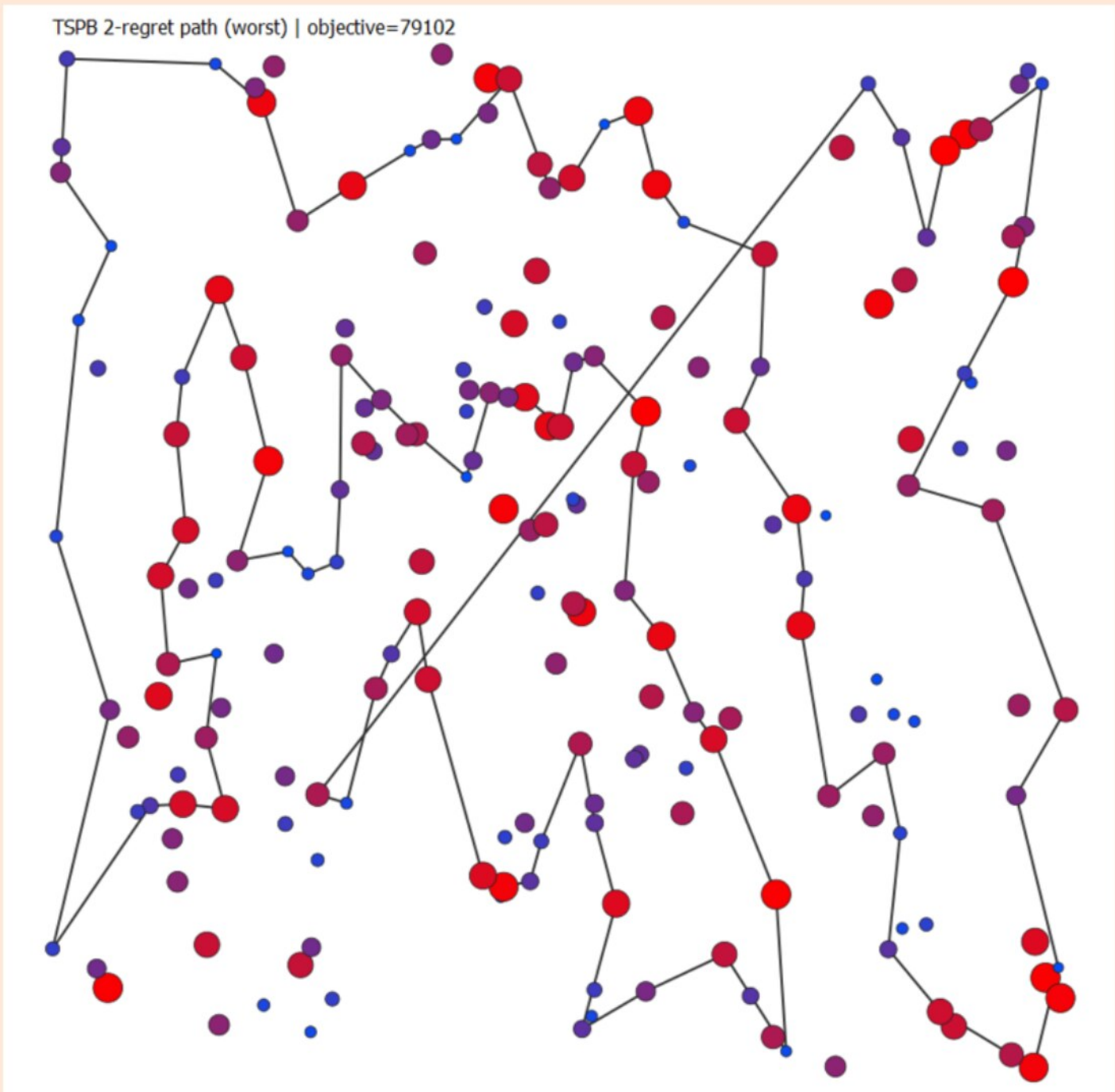
objective 76027

Worst solution nodes:

154 186 164 33 52 51 16
76 132 4 71 157 91 80 48
31 108 60 116 149 63 168
127 97 49 152 123 90 162
180 30 187 171 2 189 141
87 28 156 35 185 67 144
45 135 73 140 53 179 177
36 131 54 9 8 174 84 58
181 192 158 134 72 176
150 151 138 182 57 64 107
88 78 193 153 75 170 146
159 130 113 89 42 106 81
85 3 93 47 37 147 50 169
1 178 101 29 163 136 39
154

100 selected nodes; closed route

7. Large worst visualizations - TSPB



WORST SOLUTION

2-regret - path

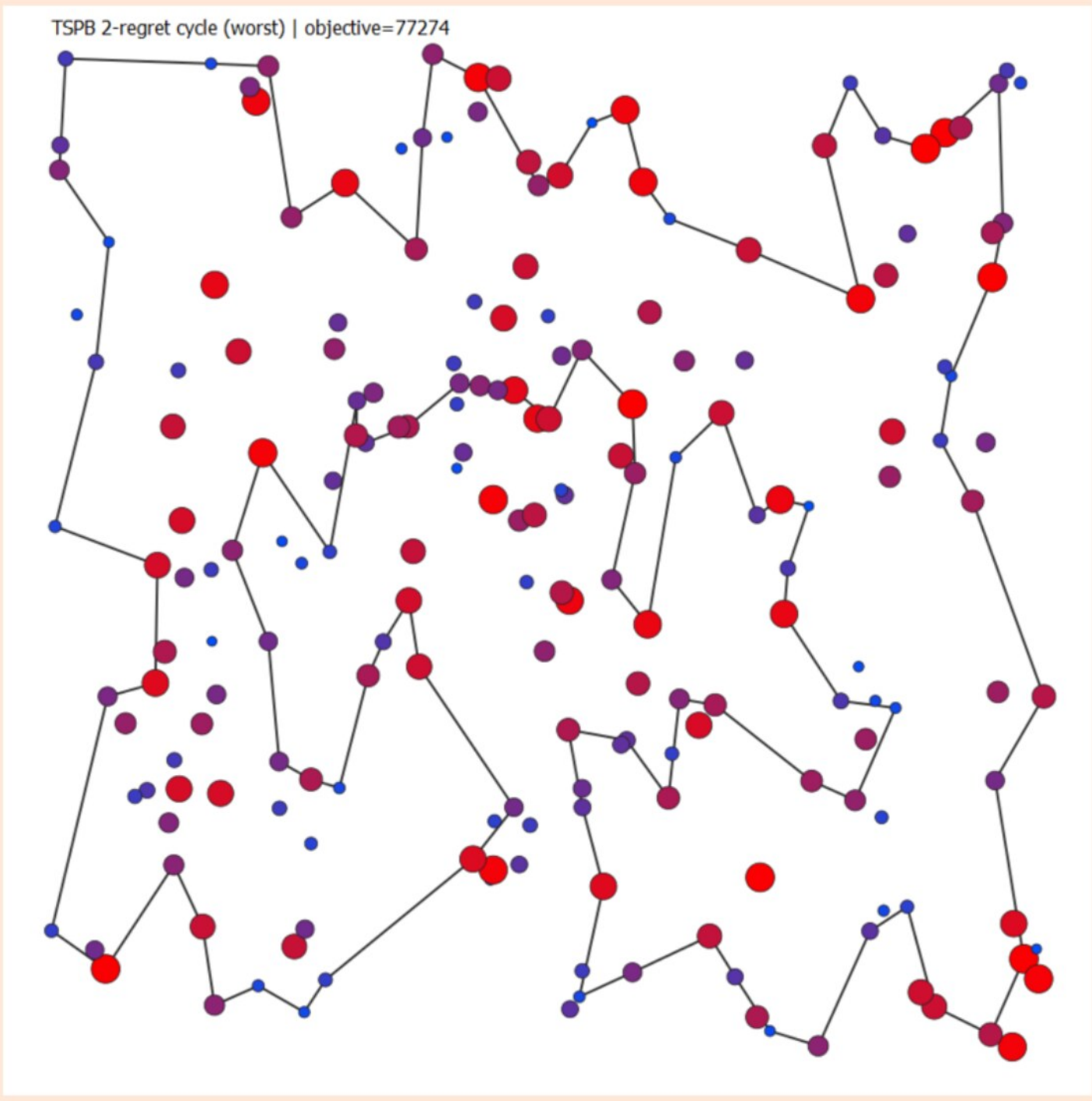
min=67937
max=79102
avg=73139.61

objective 79102

Worst solution nodes:

```
141 40 33 64 4 49 107 101
51 167 25 181 126 82 88
163 171 186 119 66 35 27
75 0 86 46 137 95 8 69
168 183 191 38 47 140 154
26 161 15 123 158 138 55
53 19 170 180 113 5 29 1
178 65 28 149 13 37 116 7
50 195 121 198 96 164 156
32 89 44 83 57 77 104 144
159 192 106 150 52 153 74
188 93 23 110 118 131 143
78 196 193 61 105 169 165
199 187 70 190 141
```

100 selected nodes; closed route



WORST SOLUTION

2-regret - cycle

min=66167
max=77274
avg=72327.22

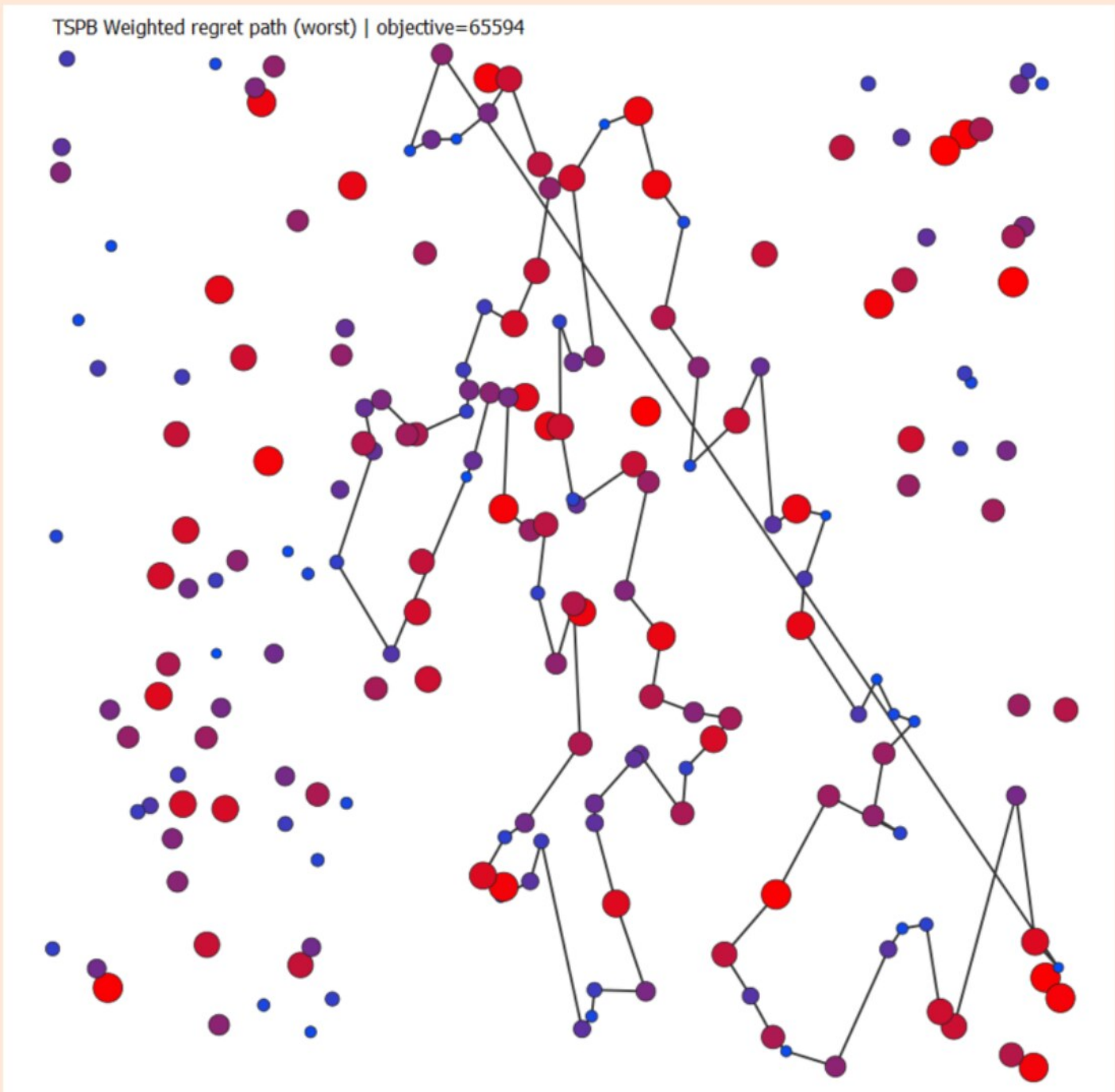
objective 77274

Worst solution nodes:

```
176 30 47 161 26 97 155
141 40 33 64 4 49 73 107
45 134 175 62 91 36 108 1
178 102 53 65 145 149 13
37 116 42 50 195 63 198
58 16 156 32 89 44 83 57
77 12 194 190 70 199 165
81 61 193 114 10 143 131
118 100 93 9 14 74 17 171
99 163 88 87 126 181 25
24 31 66 84 150 182 90 80
106 192 54 159 162 144 76
35 27 135 0 86 137 95 8
136 168 21 176
```

100 selected nodes; closed route

7. Large worst visualizations - TSPB



WORST SOLUTION

Weighted regret - path

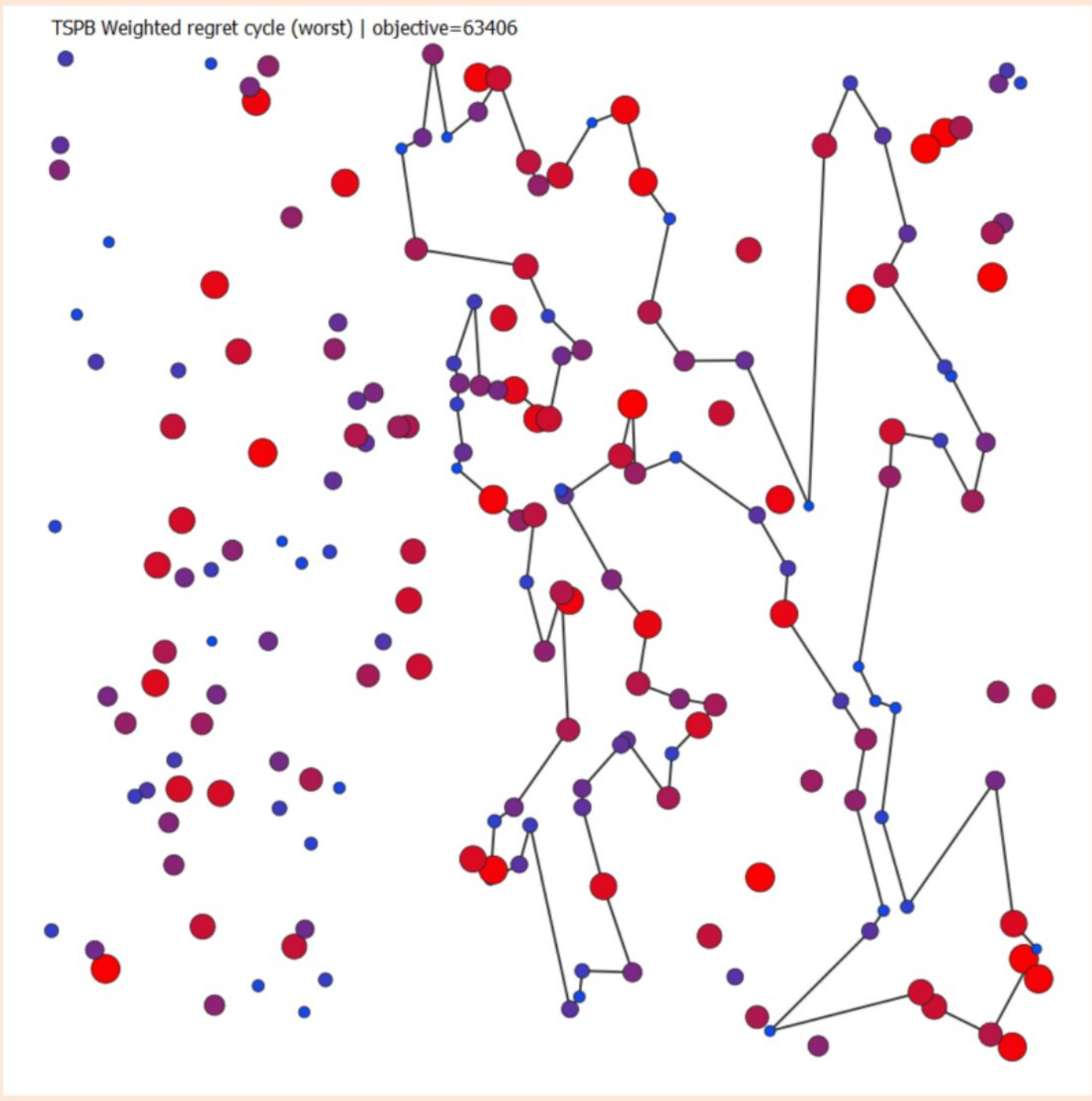
min=51301
max=65594
avg=57804.59

objective 65594

Worst solution nodes:

```
58 121 198 96 48 164 43
156 166 92 120 133 136
172 168 183 30 21 47 64
69 160 8 117 128 122 173
151 98 111 25 73 59 107
85 51 167 82 87 179 88
126 181 152 34 24 31 67
119 84 66 11 35 27 135 75
3 184 137 72 46 86 32 89
44 83 57 79 185 76 144
104 162 159 54 192 106 80
127 129 90 52 182 153 150
186 163 99 39 171 17 74
20 14 188 2 9 118 68 110
58
```

100 selected nodes; closed route



WORST SOLUTION

Weighted regret - cycle

min=52860
max=63406
avg=57903.04

objective 63406

Worst solution nodes:

```
0 135 76 162 192 106 80
52 182 20 74 171 188 2 9
93 110 68 118 14 153 90
129 127 78 41 10 143 148
114 196 18 187 70 190 194
54 104 185 79 57 83 44 89
32 156 43 164 48 96 58
198 121 63 166 72 86 46
137 95 117 8 120 133 136
172 160 69 128 122 173
151 98 111 25 73 59 85 51
167 82 87 179 88 126 181
152 34 24 31 67 119 84 66
11 35 27 184 3 75 0
```

100 selected nodes; closed route